

Appendix: A Rolling Optimization Framework for Allocating Mobile Wind Turbines in Power Distribution System Restoration

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Appendix A. Data for the Power Distribution System

The IEEE 123-node Test System is considered. The key case study parameters related to the power distribution system are listed in the following table:

Table A-I. Value of Lost Load at Each Node

Node	α_i
1	120
2	60
3	120
4	210
5	120
6	270
7	270
8	150
9	300
10	90
11	150
12	120
13	180
14	240
15	150
16	150
17	30
18	210
19	120
20	210
21	300
22	180
23	270
24	180
25	240
26	30
27	210
28	30
29	150
30	210
31	150
32	120
33	180
34	120
35	60
36	120
37	210
38	120
39	270
40	270

41	150
42	300
43	90
44	150
45	120
46	180
47	240
48	150
49	150
50	30
51	210
52	120
53	210
54	300
55	180
56	270
57	180
58	240
59	30
60	210
61	30
62	150
63	210
64	150
65	120
66	180
67	120
68	60
69	120
70	210
71	120
72	270
73	270
74	150
75	300
76	90
77	150
78	120
79	180
80	240
81	150
82	150
83	30
84	210
85	120
86	210
87	300
88	180
89	270
90	180
91	240
92	30
93	210

94	30
95	150
96	210
97	150
98	120
99	180
100	120
101	60
102	120
103	210
104	120
105	270
106	270
107	150
108	300
109	90
110	150
111	120
112	180
113	240
114	150
115	150
116	30
117	210
118	120
119	210
120	300
121	180
122	270
123	180

Table A-II. Value of Energy Not Provided by the Electric Utility at Each Time Interval

Time Interval	β_t
1	73.12
2	73.12
3	73.12
4	73.12
5	70.07
6	70.07
7	70.07
8	70.07
9	70.14
10	70.14
11	70.14
12	70.14
13	70.76
14	70.76
15	70.76
16	70.76
17	86.69
18	86.69
19	86.69
20	86.69

9	0	0	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0	0	0
11	0	0	0	0	0	0	0	0	0	0	0	0
12	0	0	0	0	0	0	0	0	0	0	0	0
13	0	0	0	0	0	0	0	0	0	0	0	0
14	0	0	0	0	0	0	0	0	0	0	0	0
15	0	0	0	0	0	0	0	0	0	0	0	0
16	0	0	0	0	0	0	0	0	0	0	0	0
17	0	0	0	0	0	0	0	0	0	0	0	0
18	0	0	0	0	0	0	0	0	0	0	0	0
19	0	0	0	0	0	0	0	0	0	0	0	0
20	0	0	0	0	0	0	0	0	0	0	0	0
21	0	0	0	0	0	0	0	0	0	0	0	0
22	0	0	0	0	0	0	0	0	0	0	0	0
23	0	0	0	0	0	0	0	0	0	0	0	0
24	0	0	0	0	0	0	0	0	0	0	0	0
25	0	0	0	1	0	0	0	0	0	0	0	1
26	0	0	0	1	0	0	0	0	0	0	0	1
27	0	1	0	1	0	0	0	0	0	0	0	1
28	0	1	0	1	0	0	0	1	0	0	0	1
29	0	1	0	1	0	0	0	1	0	1	0	1
30	0	1	0	1	0	0	0	1	0	1	0	1
31	0	1	0	1	0	0	0	1	0	1	0	1
32	0	1	0	1	1	1	0	1	0	1	0	1
33	0	1	0	1	1	1	0	1	0	1	0	1
34	0	1	0	1	1	1	0	1	0	1	0	1
35	0	1	0	1	1	1	0	1	0	1	0	1
36	0	1	0	1	1	1	1	1	0	1	0	1
37	1	1	0	1	1	1	1	1	0	1	0	1
38	1	1	0	1	1	1	1	1	0	1	0	1
39	1	1	0	1	1	1	1	1	0	1	0	1
40	1	1	0	1	1	1	1	1	0	1	0	1
41	1	1	0	1	1	1	1	1	1	1	0	1
42	1	1	0	1	1	1	1	1	1	1	0	1
43	1	1	0	1	1	1	1	1	1	1	1	1
44	1	1	1	1	1	1	1	1	1	1	1	1
45	1	1	1	1	1	1	1	1	1	1	1	1
46	1	1	1	1	1	1	1	1	1	1	1	1
47	1	1	1	1	1	1	1	1	1	1	1	1
48	1	1	1	1	1	1	1	1	1	1	1	1

Appendix B. Environment Data

The following table presents an example of wind energy availability across different segments of the power distribution system, generated through Monte Carlo simulation based on the Weibull distribution.

Table B-I. An Example of Wind Energy Availability across Different Segments

Time Interval	Segment		
	1	2	3
1	342.3299411	444.4024734	622.3921387
2	554.0940889	439.1284978	334.0258273
3	578.664764	472.4272821	587.597899
4	363.6879767	307.6008518	482.2814198

5	500.7149555	508.7285209	481.3432982
6	593.455008	428.9511436	574.0713866
7	364.4171662	378.1812656	327.5070431
8	392.1334985	220.0450435	578.5685988
9	532.5000023	363.7988184	441.798749
10	466.7508873	328.2410152	568.8365824
11	567.9709147	188.6672246	390.8607069
12	517.2949531	493.6535462	439.6855104
13	527.049358	416.1406232	578.9960106
14	410.3087843	355.5423243	457.391661
15	541.2192784	451.8577765	575.9253076
16	543.0881147	181.8640563	415.9803353
17	655.1978198	439.6992424	435.7200408
18	586.705662	309.4008161	563.176151
19	596.0314113	434.9639585	467.4815063
20	528.9657597	433.6108031	329.2749927
21	375.5940771	482.5793384	320.7277557
22	377.9855047	439.9922079	398.9508126
23	455.2771347	390.5098879	538.546047
24	509.7557443	247.1898434	300.2968617
25	421.8029621	334.2985283	451.3587586
26	429.405126	473.1171069	409.1703167
27	485.4475629	262.8389919	409.443906
28	391.9483986	164.5861386	396.3625382
29	484.4904808	589.6992424	332.1919586
30	402.504373	459.4008161	470.8799346
31	336.6991121	584.9639585	405.1811678
32	362.2413154	583.6108031	224.1838645
33	396.3761474	632.5793384	238.507683
34	585.7960465	589.9922079	246.6187244
35	427.2727428	540.5098879	271.7543593
36	516.8498749	397.1898434	304.6290403
37	593.2910487	484.2985283	350.1071588
38	456.6554145	623.1171069	179.9674404
39	443.7705883	412.8389919	242.3768457
40	502.1945942	314.5861386	268.3764002
41	379.692055	618.4093362	196.9772359
42	622.7310269	491.9266705	488.9865113
43	657.3938499	562.1674792	372.1851115
44	567.8625318	370.3523981	199.9748619
45	436.5815347	530.8676045	424.3513607
46	399.5399529	365.7410894	264.1693801
47	460.1717148	567.0333796	376.2871852
48	472.8030171	534.7639718	311.6085797